

AI safety must move from prompt rules to runtime action control.

As agent autonomy increases, the safety unit is no longer the answer. It is whether each tool call is aligned with the current intent, authority, context, and egress boundary.

NORTH STAR

Agent가 현재 환경의 의도와 aligned된 방향으로만 tool을 사용할 수 있어야 한다.

PROBLEM



- 잘못된 / 의도되지 않은 방향으로 tool이 사용됨
- Context pollution이 발생하고 safety boundary가 무너짐
- Enterprise에서는 한 번의 miss가 큰 blast radius로 확장됨

WHY EXISTING GUARDRAILS ARE NOT ENOUGH

- System prompt는 행동 경로를 추적하지 못함
- Tool description은 권한·출처·수신자 맥락을 알지 못함
- Tool-level autonomy가 커질수록 runtime decision layer가 필요함

SCOPETRACE RUNTIME LAYER

INTERCEPT EVERY TOOL CALL

1. Tool Call

agent가 행동을 요청하는 순간 캡처

2. ScopeTrace

intent, source, tool, recipient context를 결합

3. Decision

hard gate 또는 semantic review로 판단

4. Tool Result

allow/block/review 결과와 evidence를 기록

Hard Gate

반드시 지켜져야 하는 deterministic rule. 외부 입력이 privileged retrieval, CRM mutation, memory write, sensitive egress를 유도하면 즉시 차단.

Semantic Decision

명확히 금지되지는 않았지만 맥락 판단이 필요한 경우. Evidence Packet과 precedent를 기반으로 review 또는 policy proposal 생성.

ALLOW

BLOCK

REVIEW

SELF SCOPETRACE UPDATE LOOP

1

RedTeam Running Stack

Attack Planner + Specialist Attackers generate multi-turn tool-use attacks.

2

Evidence Packet

Trace, finding, source context, policy reason, safe alternative.

3

Hard Gate Proposal

Candidate rule or precedent, not active until approved.

4

Human Approval

Approve / reject / revise before runtime enforcement.

5

Replay Proof

Same attack must be replayed after patch to prove the guardrail works.

TECHNICAL STACK

Creative Operation Stack

RedTeam Running Stack

Auto-Governing Decision Maker

The unit of AI safety is no longer the answer. It is the authorized action.

ScopeTrace turns every agent action into a traceable, enforceable, and replayable governance decision.